

Scikit Learn (sklearn)

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Computational Intelligence Class

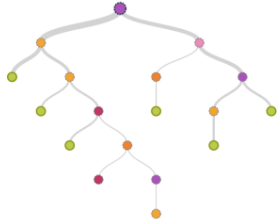
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Learning algorithms (most support classification and regression)

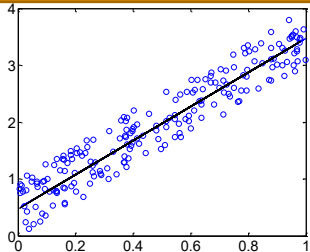
Decision stump (single if/else rule)



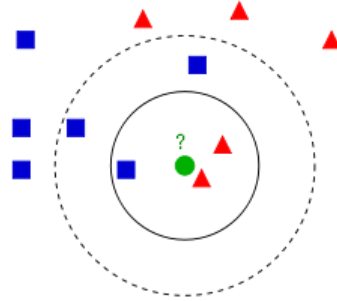
Decision tree



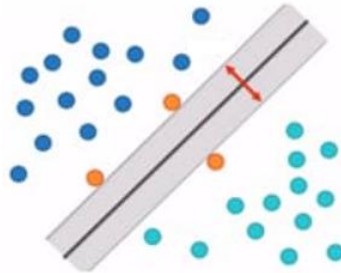
Linear regression



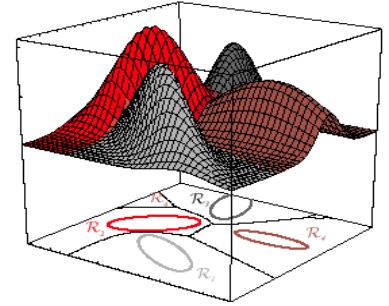
K-nearest neighbors (KNN)



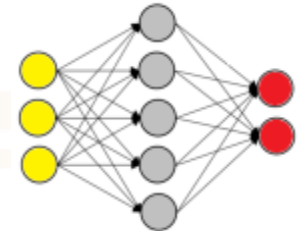
Support vector machine (SVM)



Naïve Bayes



Artificial neural network (ANN)



Scikit-learn algorithms for classification



Decision tree:

```
DecisionTreeClassifier(min_samples_leaf=5)
```

Decision stump:

```
DecisionTreeClassifier(max_depth=1)
```

Naïve Bayes

```
GaussianNB(priors=None, var_smoothing=1e-09)
```

K-nearest neighbors (KNN)

```
KNeighborsClassifier(n_neighbors=3, metric='euclidean')
```

Support vector machine (SVM)

```
SVC(C=1.0, kernel='rbf', decision_function_shape='ovo')
```

Linear (SVM)

```
LinearSVC(C=1.0, decision_function_shape='ovr')
```

Artificial neural network (ANN)

```
MLPClassifier(max_iter=500, hidden_layer_sizes=(100,))
```

Adaboost

```
AdaBoostClassifier(base_estimator=DecisionTreeClassifier(max_depth=1), n_estimators=50)
```

Random forest

```
RandomForestClassifier(n_estimators=30, max_depth=100)
```

Relevant
hyperparameters

Classification with scikit-learn



```
classifier = DecisionTreeClassifier(max_depth=20) #single tree
classifier = RandomForestClassifier(n_estimators=30,max_depth=10) #30 trees
classifier = SVC(gamma=1, C=1) #SVM with RBF kernel

classifier.fit(X_train, y_train) #training stage
y_predicted = classifier.predict(X_test) #test stage
print('Accuracy = ', accuracy_score(y_test, y_predicted))
```

Be aware that modern ML has sophisticated workflows that require time to set up and get familiar with

Advanced example

- § How to create a set of models and compare easily (Aldebaro can provide code)